Federal Research and Development Tax Credit Study

EC Malone Corporation

2021

1) Executive Summary

EC Malone Corporation ("the Company") engaged Occams Advisory ("Occams Advisory") to perform a Research and Development ("R&D") Tax Credit Study ("Study"). The Study was undertaken to analyze and document expenditures and business activities and determine qualified research expenses that may be applied towards the tax credit following the guidance of Internal Revenue Code ("IRC" or the "Code") Sections 41 and 174.

The Study covered the activities undertaken and the expenses incurred by EC Malone Corporation during the tax year 2021 (the "Study Period"). The results of the Study yielded $31,859.71 in Federal R&D tax credits.

In order to analyze, substantiate and document the Company's various R&D activities in accordance with the Code, associated regulations, and various caselaw, the Company collected the following information:

- General demographic information related to the Company's formation and qualification for the R&D tax credit;
- Federal and State (if applicable) tax return information;
- Employee list inclusive of job titles for the Study Period;
- Employee payroll amounts (W-2 box 1 wage equivalent) for each employee for the Study Period;
- A listing of qualified supplies, contractors, and cloud computing expenses (if applicable);
- A listing of projects completed during the Study Period;
- Time sheet data and/or estimates of qualified hours and activities; and
- In-depth project information related to qualified research activities.

Following the collection of the above information, Occams Advisory and/or its agents analyzed the data to ensure that the work completed, along with the associated expenditures qualified for the R&D tax credit by meeting the four-part-test outlined in the Code. Furthermore, Occams Advisory and/or its agents analyzed the scope of work and/or associated contractual agreements to determine if any projects would fall under the funded research exclusion of the Code or any other associated exclusions.

a) Summary of R&D Expenditures

	2021
Qualified Wages (See Appendix A)	$510,710.30
Total Qualified Expenses	$510,710.30
Total Federal Credit	$31,859.71

b) Summary of R&D Projects

Projects were conducted by the following employees: Jay Bruner, George Cox, Joseph Fullen, David Allen, Daniel Edwards, William Ray, Nicholas Newman, Michael Summers, Jeffery Phillipson, Troy Johnson, Christopher Murphy, Culley Fullen, Mitchell Havens, Rodney Taylor, Justin Harris, and Roman Malone. A summary of the projects analyzed during the Study along with their qualifying status is contained in the table below.

Project Name	Qualification Status
Von Braun Center	Qualified
Bishop State Community College	Qualified
Vestavia Hills Community Center	Qualified
Software Development	Qualified
Alabama School of Cyber Tech & Eng	Qualified

2) Overview of Statutory Authority

a) Qualified Research

IRC § 41 provides a tax credit for conducting research activities undertaken to develop or improve a business component. Research constitutes "qualified research" under § 41(d)(1) if it is research:

(1) With respect to which expenditures may be treated as expenses under §174;

(2) That is undertaken for the purpose of discovering information that is technological in nature;

The application of which is intended to be useful in the development of a new or improved business component of the taxpayer; and

(3) Substantially all of the activities of which constitute elements of a process of experimentation that relates to a qualified purpose.

Under §174 qualified expenses are amounts paid or incurred by the taxpayer in the taxable year in connection with his trade or business as expenses which are applied towards research or experimental activity. Expenses qualify as research or experimental costs when they are incurred in the effort to resolve uncertainty in the development or improvement of a business component. Uncertainty exists if the information available to the taxpayer does not establish the capability or method for developing or improving the business component, or the appropriate design for the business component.

To satisfy the second requirement of the statute, the research must be undertaken to discover information rooted in the principles of science. This component does not require a taxpayer seeking to obtain information that exceeds, expands or refines the common knowledge of skilled professionals in a particular field of science; nor does the component require a taxpayer to succeed in the research.

The third requirement is that the application of the research be applied to the development of a new or improved business component of the taxpayer. A business component is a product, process, formula, invention, software, or technique produced in the scope of the taxpayer's trade.

Regarding the last requirement, substantially all of a taxpayer's research activities will be considered qualified research if 80 percent or more of the research activities constitute elements of a process of experimentation that relates to a qualified purpose. A process of experimentation is a process designed to evaluate one or more alternatives to achieve a result where the capability or the method of achieving that result, or the appropriate design of that result, is uncertain as of the beginning of the taxpayer's research activities. A process of experimentation must fundamentally rely on the principles of science and involve the identification of uncertainty concerning development or improvement of a business component, the identification of one or more alternatives intended to eliminate that uncertainty, and the identification and the conduct of a process of evaluating alternatives (through, for example, modeling, simulation, or a systematic process of trial and error). This process of experimentation is undertaken for a qualified purpose if it relates to a new or improved function, performance, reliability or quality of the business component.

b) Excluded Activities

IRC § 41(d) (4) outlines activities for which the tax credit is not allowed. The excluded activities that do not meet the definition of "qualified research" are as follows:

(1) Any research conducted after the beginning of commercial production of the business component.

(2) Any research related to the adaptation of an existing business component to a particular customer's requirement or need.

(3) Any research related to the reproduction of an existing business component (in whole or in part) from a physical examination of the business component itself or from plans, blueprints, detailed specifications, or publicly available information with respect to such business component.

(4) Any –

(a) efficiency survey,

(b) activity relating to management function or technique,

(c) marketing research, testing, or development (including advertising or promotions),

(d) routine data collection, or

(e) routine or ordinary testing or inspection for quality control.

(5) Except to the extent provided in regulations, any research with respect to computer software which is developed by (or for the benefit of) the taxpayer primarily for internal use by the taxpayer, other than for use in –

(a) an activity which constitutes qualified research (determined with regard to this subparagraph), or

(b) a production process with respect to which the requirements of paragraph (1) are met.

(6) Any research conducted outside of the United States, the Commonwealth of Puerto Rico, or any possession of the United States.

(7) Any research in the social sciences, arts, or humanities.

(8) Any research to the extent funded by any grant, contract, or otherwise by another person (or governmental entity).

c) Qualified Research Expenses

IRC § 41(b)(1) explains that the term "qualified research expenses" means the sum of in-house research expenses and contract research expenses. Additionally, IRC § 41(b)(2)(A) further defines in-house research expenses as:

(1) Any wages paid or incurred to an employee for qualified services performed by such employee; and

(2) Any amount paid or incurred for supplies used in the conduct of qualified research.

Furthermore, IRC § 41(b)(3)(A) defines contract research expenses as 65 percent of any amount paid or incurred by the taxpayer to any person (other than an employee of the taxpayer) for qualified research.

i) Wages

Qualified wages are those amounts paid or incurred to an employee for qualified services performed by such employee. IRC § 41(b)(2)(B) provides that the term "qualified services" means services consisting of – 1) engaging in qualified research, or 2) engaging in the direct supervision or direct support of research activities which constitute qualified research. Furthermore, if substantially all of the services performed by an individual for the taxpayer during the taxable year consists of services meeting the requirements of clause 1 or 2, the term "qualified services" means all of the services performed by such an individual for the taxpayer during the taxable year.

Treasury Regulations § 1.41-2(d) expands upon the topic of qualified wages with the following:

> Wages paid to or incurred for an employee constitute in-house research expenses only to the extent the wages were paid or incurred for qualified services performed by the employee. If an employee has performed both qualified services and nonqualified services, only the amount of wages allocated to the performance of qualified services constitutes an in- house research expense. In the absence of another method of allocation that the taxpayer can demonstrate to be more appropriate, the amount of in-house research expense shall be determined by multiplying the total amount of wages paid or incurred for the employee during the taxable year by the ratio of the total time actually spent by the employee in the performance of qualified services for the taxpayer to the total time spent by the employee in the performance of all services for the taxpayer during the taxable year.

ii) Supplies

IRC § 41(b)(2)(B) defines "supplies" as any tangible property other than 1) land or improvements to land, and 2) property of a character subject to the allowance for depreciation. Clause 2 is not satisfied unless the property is depreciable in the hands of the taxpayer (See TG Missouri Corp. v. Commissioner, 133 TC 278, 292 (2009)).

Supply expenses are properly claimed when used in the conduct of qualified research by the taxpayer. Treasury Regulations § 1.41-2(b) (1) explains that expenditures for supplies or for the use of personal property that are indirect research expenditures or general and administrative expenses do not qualify as in- house research expenses.

iii) Contract Research Expenses

IRC § 41(b)(3)(A) defines "contract research expenses" as 65 percent of any amount paid or incurred by the taxpayer to any person (other than an employee of the taxpayer for qualified research). Essentially, contract research expenses are amounts expended for qualified services performed by a third party.

3) R&D Tax Credit Analysis by Project

i) Basic data

Project Name: Von Braun Center

Project Years: 2020 through 2021

Man hours: 2000

Contract Type: Lump sum/fixed fee

Employees Performing Research: Daniel Edwards

ii) Project Description

This project involves the design and installation of metal wall panels for the complete renovation of the Von Braun Center in Huntsville, AL.

ECM Corp was responsible for designing the wall panel configuration and fabricating the custom metal wall panels required for the spaces.

The project also included the installation and fittings of the metal wall panels, including custom fabricated Aluminum Composite Material (ACM) wall panels.

iii) Why This is a New or Improved Business Component

This project introduces novel or enhanced business components targeting performance, functionality, quality, or reliability in the following ways:

1) Optimizing Fabrication Processes for Precision Metal Wall Panels

- **Technique:** The activity 'Optimizing Fabrication Processes for Precision Metal Wall Panels' represent a new or improved technique targeting performance, functionality, quality, and reliability. By employing advanced computer modeling and software applications to ensure precise measurements and design conformity, the team enhances the accuracy and efficiency of the fabrication process. This technique significantly improves the quality and reliability of the metal wall panels, addressing the technical uncertainties related to precise dimensions and tight tolerances.

iv) Why This Project Involves the Elimination of Uncertainty (Information Sought to Discover)

There is uncertainty regarding the capability, method, or design of the following technical challenges.

1) Optimizing Fabrication Processes for Precision Metal Wall Panels

Appropriate Design: There is uncertainty related to the appropriate design of the metal wall panels due to unforeseen alterations during installation. Variances found during the installation phase require redrawing of plans, remodeling of panels, and slight alterations to the design build, which can impact the project's efficiency and quality.

v) Why This Project Involves a Process of Experimentation

1) Optimizing Fabrication Processes for Precision Metal Wall Panels

The project of optimizing fabrication processes for precision metal wall panels at the Von Braun Center involved significant uncertainty regarding the design and optimization of fabrication processes to achieve precise dimensions and tight tolerances. This uncertainty was addressed through a systematic process of experimentation involving multiple stages of measurement, computer modeling, and design iterations. Initially, the team conducted rough measurements and designs for estimation and bidding purposes. Upon winning the project, they performed very precise measurements that significantly altered the initial designs. These precise measurements were crucial to ensuring that the final designs met the exact specifications required for the metal wall panels. The use of computer modeling and software applications allowed the team to simulate various design scenarios and identify the most efficient use of materials, thereby eliminating uncertainties related to design conformity and material optimization.

To tackle the uncertainty associated with achieving precise cuts, welds, and fabrications, the team relied on computer software to model these processes. This step involved systematic trial and error, as the software simulated different fabrication methods to determine the most efficient and precise approach. The computer models helped in the entire fabrication process, from cutting to welding and assembly, ensuring that each step was optimized for precision and efficiency. By iterating through various design alternatives and evaluating their outcomes through simulations, the team was able to refine their fabrication processes to achieve the desired tight tolerances without compromising on quality.

During the installation phase, the project faced further uncertainties due to unforeseen alterations in the fit of the units. These variances required a process of experimentation to address the changes effectively. The team had to redraw plans and remodel panels to accommodate the discrepancies found during installation. This iterative process involved continuous feedback and adjustments to the design-build, ensuring that the final installation met the required specifications. The systematic approach to addressing installation variances through real-time adjustments and design iterations exemplifies the project's commitment to a process of experimentation aimed at resolving uncertainties and achieving the desired outcomes.

The project fundamentally relied on principles of engineering and computer science to address the uncertainties associated with the fabrication and installation of metal wall panels. The use of computer modeling and simulations for precise measurements and fabrication processes highlights the technological nature of the project. By employing advanced software applications to optimize material usage and ensure design conformity, the project demonstrated a reliance on technological innovations to achieve its goals. This approach underscores the project's alignment with the criteria for qualified research activities, as it involved a systematic process of experimentation aimed at eliminating technical uncertainties related to the development and installation of the metal wall panels.

vi) Why This Project Is Technological In Nature

1) Optimizing Fabrication Processes for Precision Metal Wall Panels

Engineering:

Optimizing Fabrication Processes for Precision Metal Wall Panels applied the principle of engineering by systematically addressing the technical uncertainties associated with the design, fabrication, and installation of metal wall panels for the Von Braun Center renovation project. Engineering principles were crucial in ensuring that the panels met precise dimensions and tight tolerances without compromising quality and efficiency.

Engineering's relevance to this activity is evident through the use of computer modeling and software applications to create detailed and accurate design plans. This approach allowed the team to transition from rough estimations to precise measurements, ensuring that the fabricated panels conformed to the specific requirements of the space. The application of engineering principles in this context involved a thorough

understanding of material properties, cutting techniques, and welding processes to achieve the desired outcomes.

The iterative process of measuring the space, updating design plans, and modeling the fabrication processes exemplifies the engineering principle of problem-solving. The team utilized advanced software to simulate the cutting, welding, and assembly of the panels, optimizing material usage and minimizing waste. This structured approach ensured that the fabrication processes were efficient and aligned with the project's quality standards.

During the installation phase, engineering principles were again applied to address variances and unforeseen alterations in the fit of the units. The ability to redraw plans, remodel panels, and make slight design alterations on-site demonstrates the practical application of engineering knowledge to adapt to real-world challenges. This flexibility and problem-solving capability are hallmarks of engineering, ensuring that the project can be completed successfully despite any deviations from the initial plans.

In summary, the activity of optimizing fabrication processes for precision metal wall panels at the Von Braun Center leveraged engineering principles extensively. From initial measurements and computer modeling to on-site adjustments and quality control, engineering played a critical role in achieving the project's objectives of precision, efficiency, and high-quality outcomes.

---

i) Basic data

Project Name: Bishop State Community College

Project Years: 2021

Man hours: 250

Contract Type: Lump sum/fixed fee

Employees Performing Research: David Allen, Nicholas Newman, George Cox, Daniel Edwards

ii) Project Description

This project involves the design and construction of aluminum composite material (ACM) walls, single skin, plate flashings, and insulation for a community college building.

The project will focus on creating durable and efficient wall systems using ACM, which is known for its strength and lightweight properties.

Single skin walls will be integrated with plate flashings to ensure proper water drainage and protection from the elements.

Additionally, insulation will be installed to enhance energy efficiency and provide a comfortable indoor environment for students and staff

iii) Why This is a New or Improved Business Component

This project introduces novel or enhanced business components targeting performance, functionality, quality, or reliability in the following ways:

1) Optimizing Fabrication Processes for Precise Wall Panels

- **Technique:** The activity 'Optimizing Fabrication Processes for Precise Wall Panels' represents a new or improved technique targeting performance, functionality, quality, and reliability. This activity involves the systematic use of computer modeling and software applications to ensure precise measurements, design conformity, and efficient material use, thereby enhancing the overall quality

and stability of the wall panels. By addressing uncertainties related to measurement accuracy, material stability, and guidelines, the project significantly improves the reliability and performance of the ACM wall systems, ensuring they meet the stringent requirements of the community college building.

iv) Why This Project Involves the Elimination of Uncertainty (Information Sought to Discover)

There is uncertainty regarding the capability, method, or design of the following technical challenges.

1) Optimizing Fabrication Processes for Precise Wall Panels

Appropriate Design: An uncertainty related to appropriate design is how varying measurements during the initial design build and estimation phases will affect the overall product. This includes understanding how deviations from initial measurements can impact the final installation and ensuring that the design can accommodate such variances without compromising quality or stability.

v) Why This Project Involves a Process of Experimentation

1) Optimizing Fabrication Processes for Precise Wall Panels

The Bishop State Community College project involved a process of experimentation to develop and optimize fabrication processes for precise wall panels. Initially, the team faced uncertainty regarding how to achieve precise dimensions and tight tolerances for the aluminum composite material (ACM) wall panels and accessories without compromising quality, and stability. To address this, the team conducted systematic trial and error by taking rough measurements of the space and creating initial design plans for estimation and bidding purposes. Once the project was awarded, very precise measurements were taken, significantly altering the initial rough design. Computer modeling and software applications were employed to ensure design conformity, allowing the team to refine their approach iteratively. This evaluative process ensured that the final design met the required specifications while optimizing material use.

The project also involved uncertainty regarding the implementation of guidelines from the buyer without compromising the stability of the walls. This was tackled through simulations and modeling, where the team used computer software to model precise cuts, welds, and fabrication of the units. By iterating through different design alternatives, the team was able to find a balance in the structural integrity. This process of experimentation involved analyzing various design iterations and refining the approach based on feedback and performance assessments, ensuring that the final product met stability criteria.

During the installation phase, the team encountered additional uncertainty related to variances in measurements that affected the overall product from the initial design build and estimation. To address this, the team conducted a process of systematic trial and error, redrawing plans, remodeling panels, and making slight alterations to the design build. This iterative process involved continuous feedback loops where the team assessed the impact of variances and adjusted their approach accordingly. By employing a rigorous process of experimentation, the team was able to adapt to on-site conditions and ensure that the final installation met the project's specifications and quality standards.

vi) Why This Project Is Technological In Nature

1) Optimizing Fabrication Processes for Precise Wall Panels

Engineering:

The activity 'Optimizing Fabrication Processes for Precise Wall Panels' applied the principle of engineering through a systematic approach that integrated scientific principles, mathematical methods, and practical knowledge to solve complex problems associated with the fabrication of ACM wall panels for Bishop State Community College.

Engineering principles were crucial in addressing the technical uncertainties related to achieving precise dimensions and tight tolerances in the fabrication process. The team utilized computer modeling and software applications to ensure design conformity and to optimize the cutting, welding, and assembly processes. This reliance on technology and precision engineering highlights the application of engineering principles to achieve high-quality, stable, and effective wall panels.

The engineering principle of problem-solving was evident in the iterative process of measurement, design, and fabrication. Initial rough designs and estimations were refined through precise measurements and computer modeling, ensuring that the final product met the required specifications. This approach exemplifies how engineering principles are applied to manage and resolve uncertainties related to material stability.

Furthermore, the activity involved continuous adaptation and problem-solving during installation. Variances discovered during installation required redrawing plans, remodeling panels, and making slight alterations to the design. This flexibility and responsiveness are key aspects of engineering, demonstrating the practical application of scientific principles and logical reasoning to ensure the successful completion of the project.

Overall, the activity 'Optimizing Fabrication Processes for Precise Wall Panels' showcased the application of engineering principles through the use of advanced modeling techniques, precise measurements, and iterative problem-solving. These principles were relevant to the activity as they provided the framework for achieving the project's goals of precision, stability, and efficiency in the construction of ACM walls for Bishop State Community College.

---

i) Basic data

Project Name: Vestavia Hills Community Center

Project Years: 2021

Man hours: 250

Contract Type: Lump sum/fixed fee

Employees Performing Research: David Allen, George Cox, Daniel Edwards, Nicholas Newman

ii) Project Description

This project involves the design, construction, and installation of walls for the Vestavia Hills Community Center building. The work includes ACM, single skin, flashings, plate, and insulation tasks.

The project aims to enhance the structural integrity of the community center. It covers the complete process from design to the final installation of the wall components.

iii) Why This is a New or Improved Business Component

This project introduces novel or enhanced business components targeting performance, functionality, quality, or reliability in the following ways:

1) Optimizing CAD-Based Architectural Precision for Construction Stability

- **Technique:** The activity of 'Optimizing CAD-Based Architectural Precision for Construction Stability' represents a new or improved technique targeting performance and reliability. By utilizing advanced CAD modeling to translate the architect's general drawings into precise construction plans, the project ensures that the wall components are built with exacting accuracy, thereby enhancing the structural stability and meeting the client's requirements. This technique improves performance and

reliability by addressing uncertainties related to measurement precision and architectural intent, resulting in a final product that aligns closely with the design specifications and structural needs of the Vestavia Hills Community Center.

iv) Why This Project Involves the Elimination of Uncertainty (Information Sought to Discover)

There is uncertainty regarding the capability, method, or design of the following technical challenges.

1) Optimizing CAD-Based Architectural Precision for Construction Stability

Appropriate Design: The variances between the architect's drawings and the final erection drawings needed to be adjusted from the initial estimation, presenting uncertainties that impact the project. The team had to create multiple sets of drawings, including erection drawings, to determine the precise cuts, welds, and materials required, which introduced uncertainties related to the appropriate design.

v) Why This Project Involves a Process of Experimentation

1) Optimizing CAD-Based Architectural Precision for Construction Stability

The project involved a significant level of technical uncertainty regarding how to meet the architect's intent without sacrificing the stability of the walls. The team engaged in a systematic process of experimentation to address this uncertainty. Initially, they created a building shell in CAD for estimation purposes, which provided a preliminary framework. This was followed by incorporating the architect's drawings, which included general information and product specifications. However, the true challenge lay in adjusting these preliminary plans to ensure structural stability while meeting the precise measurements required by the client and architect. This involved multiple iterations of modifying and re-modifying dimensions in CAD to balance the design intent with structural requirements.

Another critical uncertainty was how to adjust the precise measurements to meet the client's needs. To tackle this, the team conducted physical measurements of the building and redrew the building in CAD to account for these precise measurements. This phase involved systematic trial and error, where the team continuously adjusted dimensions in CAD to ensure accuracy. Each iteration required careful analysis and recalibration to align with both the architectural intent and structural integrity. This iterative process ensured that the final design was both precise and stable, addressing the client's specifications and maintaining the project's viability.

The project also faced uncertainty regarding the variances between the architect's drawings and the final erection drawings. This required a detailed process of experimentation to identify and adjust for discrepancies that impact the project. The team created another set of drawings, known as erection drawings, for the final draft of the design. These drawings were essential for determining the precise cuts, welds, and other specifications needed for the final material list. Throughout the installation phase, modifications to the final CAD design were necessary to address unforeseen variations. This ongoing process of refinement and adjustment underscored the project's reliance on systematic trial and error, simulations, and iterative design to achieve the desired outcome.

vi) Why This Project Is Technological In Nature

1) Optimizing CAD-Based Architectural Precision for Construction Stability

Engineering:

The activity 'Optimizing CAD-Based Architectural Precision for Construction Stability' applies the principle of engineering through the systematic use of computer-aided design (CAD) software to ensure the structural integrity and precision of the Vestavia Hills Community Center walls. Engineering principles are crucial in addressing the technical uncertainties related to meeting the architect's intent without sacrificing stability. The team must employ scientific principles, mathematical methods, and practical knowledge to create accurate CAD models that reflect the building's real-world measurements and conditions.

The relevance of engineering to this activity is highlighted by the iterative process of creating, modifying, and refining CAD drawings. Initially, the team creates a building shell in CAD for estimation purposes. They then incorporate the architect's general drawings, which specify the products to be installed, and begin to factor in the architect's intent. This phase involves significant problem-solving and design creativity, as the team must adjust for precise measurements to meet the client's needs while ensuring structural stability.

Engineering principles are further applied when the team physically measures the building and redraws it in CAD to account for exact dimensions. This step is critical for modifying and re-modifying dimensions to align with client and architect requirements. The creation of erection drawings for the final draft of the design demonstrates the engineering focus on precision and accuracy. These drawings are essential for determining the precise cuts, welds, and other details needed for the final material list, ensuring that all components fit together seamlessly during construction.

During installation, the need for further modifications to the final CAD design when unforeseen variations arise underscores the engineering principle of adaptability and continuous improvement. The team must be prepared to make real-time adjustments to maintain the structural integrity of the community center. This iterative and adaptive process exemplifies the engineering approach to problem-solving, where scientific analysis, innovative design, and practical implementation are integrated to address technical uncertainties and achieve the project's goals.

---

i) Basic data

Project Name: Software Development

Project Years: 2021

Man hours: 200

Contract Type: Lump sum/fixed fee

Employees Performing Research: Daniel Edwards

ii) Project Description

Our project involves developing custom software tailored for various tasks related to wall jobs throughout the organization. We have outsourced the primary development work to a team in India.

The objective is to create efficient and user-friendly software solutions that streamline our processes and improve overall productivity. This collaboration aims to leverage the expertise and effectiveness of our development partners.

iii) Why This is a New or Improved Business Component

This project introduces novel or enhanced business components targeting performance, functionality, quality, or reliability in the following ways:

1) Agile Software Development Team for Custom Application

- **Software:** The activity of Agile Software Development Team for Custom Application is a new or improved business component targeting performance, functionality, quality, and reliability. By scoping teams to find the best development support, employing agile software development methodologies, and having a supervisory team member drive the direction, the project aims to create efficient, user-friendly software solutions tailored to the organization's specific needs. This approach enhances the software's performance and functionality, ensuring a high-quality, reliable product that meets the organization's requirements.

iv) Why This Project Involves the Elimination of Uncertainty (Information Sought to Discover)

There is uncertainty regarding the capability, method, or design of the following technical challenges.

1) Agile Software Development Team for Custom Application

Appropriate Design: There was uncertainty about which design elements were more important for the application. The team needed to prioritize aspects such as user interface design, user experience, system architecture, and security features to ensure the software was both efficient and user-friendly.

Appropriate Design: Determining the required functionality for the software posed significant uncertainty. The project team needed to identify and define the features and capabilities that best support the organization's tasks related to wall jobs, ensuring that the software effectively streamlined processes and improved productivity.

v) Why This Project Involves a Process of Experimentation

1) Agile Software Development Team for Custom Application

A significant uncertainty was deciding on the programming language for the application. This decision required a thorough examination of various alternatives, considering factors such as the project's technical requirements, the development team's proficiency, and the long-term maintainability of the software. The team conducted simulations and modeling to predict the performance and scalability of different programming languages in the context of the application's intended use. Through iterative testing and evaluation, the team was able to determine the most suitable language that optimizes the software's functionality and performance, thereby eliminating the uncertainty surrounding this critical technical decision.

The design elements of the software presented another area of uncertainty. The team needed to identify which design aspects were most important to ensure the software was user-friendly and efficient. This was addressed through a process of experimentation that involved creating multiple design prototypes and conducting user testing sessions to gather feedback. The team analyzed the usability and effectiveness of each design iteration, making adjustments based on user input and performance metrics. This iterative design process, grounded in systematic trial and error, allowed the team to refine the software's interface and user experience, ultimately achieving a design that met the project's objectives.

The functionality required for the software's usage was also uncertain at the outset. The team engaged in a detailed requirements-gathering process, involving stakeholders from various departments to understand their specific needs and expectations. This process included developing use cases, conducting surveys, and holding workshops to identify the key functionalities that the software needed to support. The team then conducted a series of development sprints, each followed by testing and feedback sessions, to iteratively build and refine the software's features. This systematic approach ensured that the final product was tailored to the organization's needs and effectively addressed the initial uncertainty regarding required functionality.

vi) Why This Project Is Technological In Nature

1) Agile Software Development Team for Custom Application

Computer Science:

The activity 'Agile Software Development Team for Custom Application' applied the principles of computer science by leveraging systematic evaluation techniques, software engineering practices, and data management methodologies to address the project's uncertainties and achieve its objectives.

The relevance of computer science principles to this activity is evident in the project's scope and scale, which involved designing and architecting a custom software solution that balances the needs of individual

tasks with the organizational need to streamline processes and improve productivity. The project management approach employed, including the creation and management of the project timeline, deliverables, and risk mitigation strategies, aligns with software engineering practices. Additionally, the exploration of various development teams and programming languages demonstrates the application of computer science principles to optimize resource allocation and ensure successful project completion.

The activity also involved experimenting with various techniques and methodologies to determine the best design elements and functionalities for the custom software. This process of experimentation is a fundamental aspect of computer science. The iterative nature of agile software development, which involves continuous testing, feedback, and refinement, further highlights the application of computer science principles. These efforts required the use of algorithms, data structures, and software development practices to create efficient and effective solutions that address the complex requirements of the project.

In summary, the activity 'Agile Software Development Team for Custom Application' applied computer science principles by leveraging systematic evaluation techniques, software engineering practices, and data management methodologies to address the project's uncertainties and achieve its objectives. These principles were crucial in ensuring the robustness, interoperability, and reliability of the custom software, ultimately providing a competitive advantage and improving the quality and stability of the organization's processes.

---

i) Basic data

Project Name: Alabama School of Cyber Tech & Eng

Project Years: 2021

Man hours: 300

Contract Type: Lump sum/fixed fee

Employees Performing Research: David Allen, George Cox, Daniel Edwards, Nicholas Newman

ii) Project Description

This project focuses on designing and constructing walls for the Alabama School of Cyber Technology & Engineering. The design includes various materials such as ACM, single skin, terracotta, plate, and flashing.

ACM, or Aluminum Composite Material, is known for its durability. It will be used to create a modern and sleek exterior for the school's buildings.

Single skin walls offer an effective solution while maintaining structural integrity. These walls will provide functionality.

Terracotta is chosen for its natural thermal properties. It helps in maintaining a stable indoor climate.

Plate and flashing are essential components for waterproofing and protecting the structure. They will ensure the longevity and safety of the walls by preventing water ingress and damage.

iii) Why This is a New or Improved Business Component

This project introduces novel or enhanced business components targeting performance, functionality, quality, or reliability in the following ways:

1) Optimizing Architectural Precision and Stability Through Iterative CAD Design

- **Technique:** The activity 'Optimizing Architectural Precision and Stability Through Iterative CAD Design' represents a new or improved technique by targeting performance, functionality, and

reliability. By systematically refining the CAD models through iterative measurements and adjustments, the project enhances the precision and stability of the wall design, ensuring it meets the architect's intent without compromising structural integrity. This technique significantly improves the building's performance and reliability by addressing the variances between initial architectural drawings and final erection drawings, thereby optimizing the final material list and ensuring precise installation. This results in a stable, and viable wall construction for the Alabama School of Cyber Technology & Engineering.

iv) Why This Project Involves the Elimination of Uncertainty (Information Sought to Discover)

There is uncertainty regarding the capability, method, or design of the following technical challenges.

1) Optimizing Architectural Precision and Stability Through Iterative CAD Design

Appropriate Design: There was uncertainty about the appropriate design adjustments needed for variances between the architect's initial drawings and the final erection drawings. These adjustments had to be identified and managed from the initial estimation phase to avoid negatively impacting the project, ensuring that the final material list and installation process were accurate and effective.

v) Why This Project Involves a Process of Experimentation

1) Optimizing Architectural Precision and Stability Through Iterative CAD Design

The project of designing and constructing walls for the Alabama School of Cyber Technology & Engineering involved substantial technical uncertainty, particularly in optimizing architectural precision and stability. The primary uncertainty was how to meet the architect's intent without sacrificing the structural stability of the walls. To address this, the team employed a systematic process of experimentation, starting with the creation of the building shell in CAD for estimation purposes. This initial step involved modeling the general structure based on the architect's drawings, which included the intended materials but lacked precise measurements. The team then physically measured the building and redrew it in CAD, adjusting and re-adjusting dimensions to align with the client's and architect's requirements. This iterative process of modifying and refining the CAD model was essential to ensure both precision and stability, involving numerous design iterations and recalculations to achieve the desired outcome.

Another significant uncertainty was how to adjust precise measurements to meet the client's needs, given the variances between the architect's drawings and the final erection drawings. This required a detailed process of experimentation, involving systematic trial and error. After the initial CAD model was created, the team had to account for precise measurements obtained from physical site inspections. These measurements were then integrated into the CAD model, necessitating multiple iterations of modifications to ensure that the design met all specified requirements. This process involved continuous analysis and recalculations to adjust dimensions accurately, ensuring that the final erection drawings were both precise and functional. The iterative nature of this process exemplified a thorough evaluative methodology aimed at eliminating uncertainties and achieving architectural precision.

The project also faced uncertainties related to potential variances between the initial estimation and the final erection drawings. To address this, the team conducted a series of simulations and modeling exercises to predict and mitigate any impact from unforeseen variations. This involved creating another set of drawings (erection drawings) for the final draft of the design, which detailed the precise cuts, welds, and other specifications required for the final material list. During the installation phase, the team had to remain flexible, making real-time modifications to the final CAD design when encountering unforeseen variations. This ongoing process of evaluation and adjustment ensured that any impact from variances was minimized, demonstrating a systematic trial-and-error approach to managing uncertainties.

Throughout the project, the team employed a rigorous process of experimentation involving simulations, modeling, calculations, analysis, and design iterations. This approach was essential to resolve the technical uncertainties related to architectural precision and stability, as well as to meet the client's needs for precise

measurements and viability. By systematically evaluating and refining various design alternatives, the team ensured that the final construction was both structurally sound and feasible. The reliance on advanced CAD modeling and iterative design processes underscores the project's commitment to a process of experimentation aimed at achieving a technologically robust and precise architectural solution.

vi) Why This Project Is Technological In Nature

1) Optimizing Architectural Precision and Stability Through Iterative CAD Design

Engineering:

The activity 'Optimizing Architectural Precision and Stability Through Iterative CAD Design' applied the principle of engineering by addressing the technical uncertainties associated with ensuring the stability and precision of the walls for the Alabama School of Cyber Technology & Engineering. Engineering principles were utilized to systematically create, modify, and refine the CAD designs to meet both the architect's intent and the structural requirements of the project.

Engineering's relevance to this activity is evident in the iterative process of creating the building shell in CAD for estimation work. This initial step required a deep understanding of material properties and structural integrity to ensure that the design was practically implemented. The team then took the architect's general drawings and began factoring in the architectural intent, which involved translating conceptual designs into precise technical specifications.

The process of physically measuring the building and redrawing it in CAD to account for precise measurements showcases the application of engineering principles. This step required the team to modify and re-modify dimensions to align with the client's and architect's requirements while maintaining structural stability. The ability to adjust for variances between the architect's drawings and the final erection drawings demonstrates the problem-solving aspect of engineering, ensuring that the final design is both accurate and feasible.

Creating the erection drawings for the final draft of the design involved detailed engineering analysis to determine the precise cuts, welds, and other specifications needed for the final material list. This step is crucial for ensuring that the construction process is efficient and that the materials used are optimized for functionality. The iterative nature of this process highlights the engineering focus on continuous improvement and precision.

During installation, the need to make modifications to the final CAD design when unforeseen variations are encountered further underscores the engineering principle of adaptability. Engineers must be able to quickly assess and address any discrepancies that arise during construction, ensuring that the final structure meets all safety and design requirements. This dynamic problem-solving approach is a hallmark of engineering practice, ensuring the successful completion of complex projects like the Alabama School of Cyber Technology & Engineering.

Appendix A: R&D Qualified Wages Calculation

		Tax Year 2021
Qualified ERC Wages	B.1	$418,093.00
Non-R&D Wages	B.2	$345,060.98
Qualified R&D Wages Excluded Due to ERC	B = B.1 - B.2	$73,032.02
Qualified R&D Wages	A	$510,710.29
Qualified R&D Wages Excluded Due to ERC	B	$73,032.02
Net Qualified R&D Wages	C = A - B	$437,678.28

Wages used to calculate the Employee Retention credit in 2021 cannot be utilized for the R&D tax credit.